

ON IDENTIFYING THE DEGREE OF COMPETITIVENESS FROM INDUSTRY PRICE AND OUTPUT DATA

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Under the assumption that the industry inverse demand and cost functions are twice continuously differentiable, the index of competitiveness of the industry λ cannot be identified from data on industry price P and output Q and other exogenous variables z_1, z_2 alone if and only if the industry inverse demand function $P = f(Q, z_1)$ is separable in z_1 but does not take the special form $P = Q^{-1/\lambda}r(z_1) + s(Q)$.

1. Statement of the problem

Let the inverse demand function for the output of an industry be given by

$$P = f(Q, z_1), \quad (1)$$

where P is the price and Q is the quantity of the output and z_1 is a vector of exogenous variables affecting the industry demand but not the marginal cost function.¹ The latter is given by

$$MC = g(Q, z_2), \quad (2)$$

where z_2 is a vector of exogenous variables affecting the industry marginal cost but not the demand function.

If the market is characterized by perfect competition, the equilibrium

¹ One can include a vector of exogenous variables affecting both the industry demand and marginal cost functions without materially affecting the analysis here.

price and quantity are determined by

$$P = f(Q, z_1) = MC = g(Q, z_2). \quad (3)$$

If the market is characterized by monopoly, the equilibrium price and quantity are determined by:

$$MR = f(Q, z_1) + \frac{\partial f}{\partial Q}(Q, z_1)Q = g(Q, z_2). \quad (4)$$

More generally, one can write

$$f(Q, z_1) + \lambda \frac{\partial f}{\partial Q}(Q, z_1)Q = g(Q, z_2), \quad (5)$$

where λ is a constant between zero (perfect competition) and unity (monopoly) which can be interpreted as an index of the degree of competitiveness of the industry.²

The question of interest, suggested by Bresnahan (1982), is: under what condition can the constant λ be identified only from data on P , Q , z_1 and z_2 , assuming that as many and as varied observations as are necessary can be obtained? Note that information on cost is assumed to be unavailable so that the industry marginal cost function is free to assume alternative arbitrary forms subject only to the condition that the P and Q reduced form functions remain invariant.

Eq. (5) defines implicitly the reduced form function of Q in terms of z_1 and z_2 , say $h_2(z_1, z_2)$. $h_2(z_1, z_2)$ is always identified. Given $h_2(z_1, z_2)$ the reduced form function of P in terms of z_1 and z_2 , say $h_1(z_1, z_2)$, is given by eq. (1):

$$P = f(Q, z_1) = f(h_2(z_1, z_2), z_1) = h_1(z_1, z_2). \quad (6)$$

$h_1(z_1, z_2)$ is also always identified. Moreover, given $Q = h_2(z_1, z_2)$, and non-trivial dependence of $h_2(z_1, z_2)$ on z_2 ,³ the P reduced form function $h_1(z_1, z_2)$ can remain invariant only if $f(\cdot)$ is invariant:

$$h_1(z_1, z_2) = f(h_2(z_1, z_2), z_1) \quad \forall z_1, z_2. \quad (7)$$

² For a justification of eq. (5) see Bresnahan (1982) and the references contained therein.

³ $h_2(z_1, z_2)$ will depend non-trivially on z_2 if $g(Q, z_2)$ depends non-trivially on z_2 .

The question of identification then becomes: given $f(Q, z_1)$ (which must remain invariant in order for the P and Q reduced forms to be invariant), is it possible to find two distinct sets of $g(Q, z_2)$ and λ satisfying eq. (5) and giving rise to the same reduced form functions?⁴

2. An impossibility theorem

We shall show that identification is impossible if and only if the inverse demand function is separable in z_1 , that is, it takes the form

$$P = f(Q, r(z_1)). \quad (8)$$

To show necessity we observe that non-identification implies

$$f(Q, z_1) + \lambda \frac{\partial f}{\partial Q}(Q, z_1)Q = g(Q, z_2) \quad \forall z_1, z_2, \text{ and} \quad (9)$$

$$f(Q, z_1) + \lambda^* \frac{\partial f}{\partial Q}(Q, z_1)Q = g^*(Q, z_2) \quad \forall z_1, z_2, \quad (10)$$

where $\lambda \neq \lambda^*$ and the reduced form functions $Q = h_2(z_1, z_2)$ and $Q = h_2^*(z_1, z_2)$ defined by eqs. (9) and (10) respectively are identical. This in turn implies

$$\begin{aligned} \frac{\partial h_2}{\partial z_1} &= - \frac{\frac{\partial f}{\partial z_1} + \lambda \frac{\partial^2 f}{\partial z_1 \partial Q} Q}{(1 + \lambda) \frac{\partial f}{\partial Q} + \lambda \frac{\partial^2 f}{\partial Q^2} Q - \frac{\partial g}{\partial Q}} = \frac{\partial h_2^*}{\partial z_1} \\ &= - \frac{\frac{\partial f}{\partial z_1} + \lambda^* \frac{\partial^2 f}{\partial z_1 \partial Q} Q}{(1 + \lambda) \frac{\partial f}{\partial Q} + \lambda^* \frac{\partial^2 f}{\partial Q^2} Q - \frac{\partial g^*}{\partial Q}}, \quad (11) \\ \frac{\partial h_2}{\partial z_2} &= - \frac{-\frac{\partial g}{\partial z_2}}{(1 + \lambda) \frac{\partial f}{\partial Q} + \lambda \frac{\partial^2 f}{\partial Q^2} Q - \frac{\partial g}{\partial Q}} = \frac{\partial h_2^*}{\partial z_2} \end{aligned}$$

⁴ It should be clear that if the Q reduced forms are identical the P reduced forms are also identical.

$$= - \frac{-\frac{\partial g^*}{\partial z_2}}{(1 + \lambda^*) \frac{\partial f}{\partial Q} + \lambda^* \frac{\partial^2 f}{\partial Q^2} Q - \frac{\partial g^*}{\partial Q}}. \quad (12)$$

If z_2 is a scalar one can always express

$$g^*(Q, z_2) = F(g(Q, z_2), Q) \quad \forall Q, z_2 \quad (13)$$

for some $F(\cdot)$. Eq. (13) implies, by eqs. (9) and (10), that

$$f(Q, z_1) + \lambda^* \frac{\partial f}{\partial Q}(Q, z_1)Q = F\left(f(Q, z_1) + \lambda \frac{\partial f}{\partial Q}(Q, z_1)Q, Q\right) \quad \forall Q, z_1. \quad (14)$$

If z_2 is not a scalar, then eq. (12) implies

$$\frac{\partial g}{\partial z_2}(Q, z_2) = \eta(Q, z_2) \frac{\partial g^*}{\partial z_2}(Q, z_2) \quad \forall Q, z_2 \quad (15)$$

which in turn implies eq. (13),⁵ leading once again to eq. (14).

If z_1 is a scalar, then for any given λ^* , the left-hand side of eq. (14) is a function of Q and z_1 . The right-hand side is a function of two independent variables, $f(Q, z_1) + (\lambda(\partial f/\partial Q))(Q, z_1)Q$ and Q . If the transformation between these two alternative sets of independent variables is non-singular, there always exists a $F(\cdot, Q)$ such that eq. (14) holds. The transformation becomes singular only if

$$\frac{\partial f}{\partial z_1}(Q, z_1) + \lambda \frac{\partial^2 f}{\partial z_1 \partial Q}(Q, z_1)Q = 0 \quad (16)$$

which can be integrated to yield

$$f(Q, z_1) = Q^{-1/\lambda} r^*(z_1) + s(Q). \quad (17)$$

⁵ For a proof of this result, see Goldman and Uzawa (1964, pp. 388–389). Note that Q depends on z_1 and z_2 .

Thus, except for this singular case, λ cannot be uniquely identified when z_1 is a scalar.

If z_1 is not a scalar, we differentiate eq. (14) with respect to z_1 , obtaining

$$\frac{\partial f}{\partial z_1}(Q, z_1) + \lambda^* \frac{\partial^2 f}{\partial z_1 \partial Q}(Q, z_1)Q = F_1 \frac{\partial f}{\partial z_1}(Q, z_1) + \lambda \frac{\partial^2 f}{\partial z_1 \partial Q}(Q, z_1)Q$$

or (18)

$$(1 - F_1) \frac{\partial f}{\partial z_1}(Q, z_1) = (\lambda F_1 - \lambda^*) \frac{\partial^2 f}{\partial z_1 \partial Q}(Q, z_1)Q, \quad (19)$$

where $\lambda^* - \lambda \neq 0$. Eq. (19) implies

$$\frac{(\partial f / \partial z_{1i})(Q, z_1)}{(\partial f / \partial z_{1j})(Q, z_1)} = \frac{\partial^2 f / \partial z_{1i} \partial Q}{\partial^2 f / \partial z_{1j} \partial Q}, \quad i \neq j, \quad \forall i, j$$

which in turn implies

$$\frac{\partial}{\partial Q} \left(\frac{(\partial f / \partial z_{1i})(Q, z_1)}{(\partial f / \partial z_{1j})(Q, z_1)} \right) = 0, \quad i \neq j, \quad \forall i, j.$$

Thus we conclude that $f(Q, z_1)$ must be separable in z_1 .

To show sufficiency we observe that if $f(Q, z_1)$ is separable in z_1 eq. (14) becomes

$$\begin{aligned} f(Q, r(z_1)) + \lambda^* \frac{\partial f}{\partial Q}(Q, r(z_1))Q \\ = F \left(f(Q, r(z_1)) + \lambda \frac{\partial f}{\partial Q}(Q, r(z_1))Q, Q \right) \quad \forall Q, z_1. \end{aligned} \quad (20)$$

However, $r(z_1)$ can be treated as a single variable. Thus, our previous analysis of the scalar z_1 case applies, which implies that λ cannot be uniquely identified except for the singular case.

Hence, we have proved:

Theorem. Under the assumption that the industry inverse demand and cost functions are twice continuously differentiable, the index of competitiveness

λ cannot be identified from data on industry price and output and other exogenous variables alone if and only if the industry inverse demand function $P = f(Q, z_1)$ is separable in Z , but does not take the form $P = Q^{-1/\lambda} r(z_1) + s(Q)$.⁶

3. Concluding remarks

We conclude that it is necessary and sufficient for identifiability that the inverse demand function is not separable in z_1 and the dimensionality of z_1 is at least two (except for the singular case). It is easy to see that the inverse demand function is separable in z_1 if and only if the demand function is separable in z_1 . This thus constitutes a rigorous restatement of what Bresnahan (1982) refers to as the necessity for interaction effects for identification. Sufficiency is also demonstrated here.

We can cite several applications of our theorem:

- (1) If the inverse demand function (and hence the demand function) is linear:

$$P = P_0 + \alpha_Q Q + \alpha' z_1$$

then it is separable in z_1 . Hence it is impossible to identify λ .

- (2) If the inverse demand function has the constant-elasticity property $f(Q, z_1) = Q^k r(z_1)$, k a constant, which is separable in z_1 . Thus it is impossible to identify λ unless $k = -1/\lambda$.
- (3) The reader can verify that in Bresnahan's (1982) example the implied inverse demand function is not separable.
- (4) Let $TR = PQ = f(Q, z_1)Q$ be the total revenue function. If the total revenue function is polynomial in Q and z_1 and takes the value zero when Q is zero, then identification is impossible unless the polynomial is at least of third degree in Q . Alternatively, the inverse demand function (also a polynomial) must be at least of the second degree in Q .

What happens if λ were not necessarily constant? If $\lambda = \lambda(Q, z_1)$ the

⁶ Note that this theorem is true only in the absence of any restriction on the form of the industry marginal cost function as long as it depends non-trivially on z_2 .

theorem still holds in the sense that there is non-identification of the function $\lambda(Q, z_1)$ if and only if $P = f(Q, r(z_1))$ and $\lambda = \lambda(Q, r(z_1))$ except for singular cases.

What happens if there were prior information on the industry marginal cost function, for example, if it were known to be constant? In that case, $g^*(Q, z_2) = k^*$, a constant. Eq. (14) becomes

$$(1/k^* - 1/k)f(Q, z_1) + (\lambda^* - \lambda)(\partial f / \partial Q)(Q, z_1)Q = 0$$

which implies that it is impossible to identify λ if and only if the inverse demand function has the constant-elasticity property with respect to the quantity of output.⁷

References

- Bresnahan, T.F., 1982, The oligopoly solution concept is identified, Mimeo., Jan. (Department of Economics, Stanford University, Stanford, CA).
 Goldman, S.M. and H. Uzawa, 1964, A note on separability in demand analysis, *Econometrica* 32, 387-398.

⁷ In general, if there is prior information restricting the possible industry marginal cost functions to a certain class, then the transformation $F(\cdot, Q)$ in eq. (14) is restricted to one which takes any member of the admissible class of marginal cost functions into another member of the admissible class.